

- 1 (a) (i) weight = GMm/r^2 C1
 $= (6.67 \times 10^{-11} \times 6.42 \times 10^{23} \times 1.40)/(\frac{1}{2} \times 6.79 \times 10^6)^2$ C1
 $= 5.20 \text{ N}$ A1 [3]
- (ii) potential energy = $-GMm/r$ C1
 $= -(6.67 \times 10^{-11} \times 6.42 \times 10^{23} \times 1.40)/(\frac{1}{2} \times 6.79 \times 10^6)$ M1
 $= -1.77 \times 10^7 \text{ J}$ A0 [2]
- (b) either $\frac{1}{2}mv^2 = 1.77 \times 10^7$ C1
 $v^2 = (1.77 \times 10^7 \times 2)/1.40$ C1
 $v = 5.03 \times 10^3 \text{ ms}^{-1}$ A1
or $\frac{1}{2}mv^2 = GMm/r$ (C1)
 $v^2 = (2 \times 6.67 \times 10^{-11} \times 6.42 \times 10^{23})/(6.79 \times 10^6/2)$ (C1)
 $v = 5.02 \times 10^3 \text{ ms}^{-1}$ (A1) [3]
- (c) (i) $\frac{1}{2} \times 2 \times 1.66 \times 10^{-27} \times (5.03 \times 10^3)^2 = \frac{3}{2} \times 1.38 \times 10^{-23} \times T$ C1
 $T = 2030 \text{ K}$ A1 [2]
- (ii) either because there is a range of speeds M1
some molecules have a higher speed A1
or some escape from point above planet surface (M1)
so initial potential energy is higher (A1) [2]

2 (a) temperature scale calibrated assuming linear change of property with